GenCC Recommendation for Gain of Function (GOF) mechanism curation - Version 1.0

The Gene Curation Coalition proposes a semi-quantitative metric to determine the mechanism of disease for gene-disease relationships. The framework is intended to be used for human gene-disease relationships that have scored at Moderate or higher using the ClinGen gene curation framework (Strande et al 2017, see here for the most recent gene curation SOP). Levels should be capped at "Suspected" if applying to genes with Limited or below evidence. Caution should be exercised if adapting to non-human genetic diseases as this framework has not been tested using non-human phenotypes.

This framework is for evaluating genes for a gain of function mechanism.

**While this is a semi-quantitative framework, please note that points are not statistical in nature, nor do they correspond to Bayesian points in the sequence variant classification frameworks.

Table 1. Scoring Matrix for Gain of Function Framework

Variant Type[1]	Evidence	Points
GOF Variant[2]	Strong functional variant-level evidence demonstrating GOF	0.15 pt (range 0.1-0.2)
Experimental Evidence	KI or transgenic model organism with consistent phenotype and zygosity[3]	0.3 pt maximum cap (range 0.1-0.3)
	Known function of the gene supports a GOF mechanism	0.1-0.3pts
	Studies consistent with the GOF mechanistic narrative[4]	0.1 pts
	KO animal model has no phenotype (or an entirely different phenotype)	0.1 pts
Variant spectrum	Absence of pLOF variants causing disease OR that pLOF variants cause a different disease[5]	0.5 pts
	Presence of >3 recurrent variants in a cluster	0.1 pts

Footnotes:
[1]Variants can only be counted when part of a case that has variants of the types above that are assumed to be causal based on consistent phenotype. Each variant can only be scored once; even if that variant has been observed in recurrent cases.
[2]GOF variants are usually (but may not be limited to) missense variants, variants that cause in-frame deletions or insertions, or truncating variants that are predicted to escape NMD.
[3]This animal model MUST demonstrate a GOF mechanism. If it does not specifically demonstrate GOF, it can be used for other experiments that demonstrate this mechanism (e.g. ex vivo experiments on tissues or cells from the organism).
[4]Examples of this would be an expression study showing new expression in new tissues or at new developmental time points, mislocalization of the protein, showing that a protein is localized or accumulates in a new tissue or cellular location, or there is now a new function of the gene.
[5]If phenotypic features for the posed LOF and GOF disease entities are overlapping, no score should be given in this category.

Abbreviations: GOF, Gain of Function; KI, Knock in; KO, knock out; pLOF, predicted LOF; NMD, nonsense-mediated decay

Table 2. Summary Matrix for Gain of Function Framework

Points from Table 1 should be summed and the equivalent confidence term applied to the GOF mechanism

Evidence Points	Confidence Term	Mechanism
.99 or higher	Established	Gain of Function
0.90-0.98	Likely	
0.5-0.89	Suspected	
0.49-0	Uncertain	